

Predictive Multiplicity in Classification

CHARLES T. MARX, Haverford College

FLAVIO P. CALMON, Harvard SEAS

BERK USTUN, Harvard SEAS

Prediction problems often admit competing models that perform almost equally well. This effect – called the *multiplicity of good models* – challenges how we build and deploy predictive models. In this paper, we study a specific notion of model multiplicity – *predictive multiplicity* – where competing models assign conflicting predictions. Predictive multiplicity signals irreconcilable differences in the predictions of competing models. In applications such as recidivism prediction and credit scoring, evidence of predictive multiplicity challenges model selection and downstream processes that depend on it. We propose measures to evaluate the severity of predictive multiplicity in classification, and develop integer programming methods to compute them efficiently. We apply our methods to evaluate predictive multiplicity in recidivism prediction problems. Our results show that real-world datasets may admit competing models that assign wildly conflicting predictions, and support the need to measure and report predictive multiplicity in model development.

Keywords: multiplicity, classification, integer programming, recidivism prediction, measurement, contestability

1 Introduction

Many machine learning algorithms are designed to learn a best-fitting model from data. In modern methods for empirical risk minimization, for example, we fit a model that optimizes a specific objective function (e.g., the error rate) from a collection of models that obeys a specific set of constraints (e.g., linear classifiers with equal FPR between groups). In an idealized setting where we are given a large dataset of representative examples, where stakeholders have agreed on the problem formulation [38, 44], and where we have the tools to solve this problem, we would still face a key issue in applying such methods: *there may be more than one best-fitting model*.

In machine learning, *multiplicity* refers to the ability of prediction problem to admit multiple “good” models that perform almost equally well. Several works mention that prediction problems can exhibit multiplicity [36, 39], but only few discuss its implications. The work of Breiman [8] is a major exception. In a seminal position paper, Breiman describes how multiplicity challenges the interpretability of predictive models: *if there exists several good models for a given dataset – each of which provides a different explanation of the data-generating process – then how can we tell which one is correct?* Drawing parallels between the discordant explanations of good models and the discordant testimonies of witnesses in the motion picture “Rashomon,” Breiman refers to this dilemma the *Rashomon effect*. In the context of his paper, the Rashomon effect is – in fact – an argument against the misuse of predictive models in science. Seeing how prediction tasks can exhibit multiplicity, one should not draw scientific conclusions by explaining a single model – at least until one can rule out potential multiplicity.

Machine learning has changed dramatically since Breiman coined the Rashomon effect. Many models are now built for the sole purpose of making accurate predictions [26, 49]. In applications such as credit

Authors’ addresses: Charles T. Marx, Haverford College, Haverford, PA, cmarx@haverford.edu; Flavio P. Calmon, Harvard SEAS, Cambridge, MA, flavio@seas.harvard.edu; Berk Ustun, Harvard SEAS, Cambridge, MA, berk@seas.harvard.edu.

scoring and recidivism prediction, predictions affect people [7]. In turn, multiplicity challenges how we build and deploy machine learning models. Consider the following examples:

Recidivism Prediction: Say we are given a recidivism prediction model that performs well, but there exists a competing model that performs equally well. If the models assign conflicting predictions, an individual who are predicted to recidivate by a model we deploy would be predicted not to recidivate by a competing model that is just as valid. In this case, competing models with conflicting predictions present *irreconcilable differences* in predictions, similar to the impossibility results in the fair machine learning literature [see e.g., 11, 12, 27, 31]. Ignoring the existence of competing models would mean that we would arbitrate conflicting predictions by brute luck. In the worst case, one could exploit this degree of freedom to choose a model that optimizes an adversarial objective (e.g., a competing model that is least accurate on a certain group).

Credit Scoring: Say we must explain the prediction that a person receives from a scoring model to a person [by producing e.g., a counterfactual explanation 33, 54]. If there exists several models that perform equally well, models with conflicting predictions lead to contradictory explanations. In turn, it would be misguided to trust the explanations of a model for a person who is assigned a conflicting prediction by a competing model. Evidence of competing models with conflicting predictions would mitigate unwarranted rationalization of the model (e.g., via *fairwashing* [2] or *explanation bias* [24, 29]).

Stakeholders in machine learning — i.e., practitioners, policy-makers, decision-subjects — are broadly unaware that there may exist competing models that assign different predictions from models in deployment. One possible reason for this is the hypothesis that “good” models are “good” *because they assign similar predictions* [see e.g., 35, for a critique]. It is important to raise awareness that competing models can assign conflicting predictions, and to develop machinery to evaluate the incidence and severity of this effect [see e.g. 28]. The latter goal is crucial since the incidence of predictive multiplicity inherently depends on decisions made in model development such as the training data and model class. Presenting stakeholders with information on this effect allows them to challenge major decisions in model development [as per the principles of contestability 21, 28] — from model selection (e.g., how should we choose between competing models, if we should choose one at all, and who should have a say in this decision) to downstream tasks that depend on model choice (e.g., post-hoc explanation).

In this paper, we introduce measures and methods to evaluate the multiplicity of predictions in classification. We aim to provide stakeholders with meaningful information about predictive multiplicity. We introduce the following measures that capture of how predictions vary over the set of competing models:

Ambiguity: How many individuals can be assigned a different prediction by at least one good model?

Discrepancy: What is the maximum number of conflicting predictions between a baseline model and any good model?

These are model-agnostic measures that capture how multiplicity affects individuals in classification problems.

To compute these measures, one must find specific models among competing models (e.g., a model that maximizes discrepancy yet performs almost as well as the baseline model we would deploy). We develop tools to compute these quantities exactly for linear classification problems. We then apply

(x_1, x_2)	# EXAMPLES		OPTIMAL CLASSIFIERS					
	n^+	n^-	$\hat{h}_0$	$\hat{h}_a$	$\hat{h}_b$	$\hat{h}_c$		
(0, 0)	0	100	–	–	–	+	Training Error of Optimal Classifier	25%
(0, 1)	100	0	+	–	+	+	Test Error of Optimal Classifier	25%
(1, 0)	100	0	+	+	–	+	Discrepancy among Optimal Classifiers	50%
(1, 1)	0	100	+	–	–	–	Ambiguity among Optimal Classifiers	100%

Fig. 1. Example showing how classifiers that perform equally well may assign conflicting predictions. We fit linear classifiers on perfectly sampled dataset with 400 points. Here: n^+ and n^- denote the number of examples with $y = +1$ and $y = -1$ for each $(x_1, x_2) \in \{0, 1\}^2$; $\hat{h}_0, \hat{h}_a, \hat{h}_b, \hat{h}_c$ represent linear classifiers that minimize training and test error (25%). The predictions of any 2 classifiers differ on 200 points (50% of the training sample). Thus, the discrepancy is 50% among the set of optimal classifiers. There exists a classifier that can predict +1 or -1 for each point. Thus, the ambiguity is 100% among optimal classifiers.

them to measure predictive multiplicity on recidivism prediction tasks where predictive multiplicity has important ethical implications.

Contributions. The main contributions of this paper are:

1. We propose formal measures of how multiplicity affects individual predictions in a classification task: *discrepancy* and *ambiguity*.
2. We develop integer programming tools to compute these measures for linear classification problems in a way that is exact and efficient. Computing these measures requires solving hard optimization problems over a user-defined set of good models. We present integer programming formulations that recover the certifiably optimal solutions to these problems, and pair them with *path algorithms* to efficiently compute these measures over all possible sets of good models.
3. We present an empirical study of predictive multiplicity in recidivism prediction. Our results show that real-world datasets can admit good models with wildly conflicting predictions. For example, in the ProPublica COMPAS dataset [4], we find that a competing model that is only 1% less accurate than the most accurate model we can deploy, can disagree on over 17% of predictions, and that 44% of predictions are vulnerable to model selection (see Section 4).
4. We discuss how predictive multiplicity challenges key practices in machine learning. We show how our tools can calibrate trust in explanations, or address performance disparities between protected groups.
5. We provide a software implementation of our tools in Python at <https://github.com/charliemarx/pmtools>

1.1 Related Work

Multiplicity. The potential for predictive models to exhibit multiplicity has been known since at least the 1980s [see e.g., 39]. McCullagh and Nelder [36], for example, write “data will often point with almost equal emphasis on several possible models, and it is important... [to] recognize and accept this.”

Breiman [8] presents an early discussion of the implications of multiplicity. He argues that the existence of alternative models undermines the use of a single model to support broader claims about the data-generating mechanism (e.g., conflating the important variables of a recidivism prediction instrument with the root causes of criminal behavior). Breiman does not propose a solution for multiplicity. Instead, he cautions against “automatic methods of model selection” and warns that “the multiplicity problem and its effect on conclusions drawn from models needs serious attention.”

Recent work in machine learning tackles multiplicity from the “Rashomon” perspective. Fisher et al. [15] and Dong and Rudin [14] present methods to measure variable importance over the set of good models. Semenova and Rudin [48] present a measure for the relative size of the set of good models and use it to characterize settings where simple models perform well. Our work differs from this stream of research in that we focus on problems where good models output conflicting predictions (see e.g., Figure 2). This type of multiplicity leads to irreconcilable differences between subsets of predictions – similar to the impossibility results in the fair machine learning literature [see e.g., 11, 12, 27, 31].

Model Selection. Techniques to resolve multiplicity typically consist of techniques for tie-breaking or reconciliation. Classical approaches for model selection aim to choose a single model that is “closest” to a “true model” by breaking ties on the basis of estimators such as the AIC, BIC, or K-CV error [see e.g., 35, for a critique]. More recent approaches propose choosing a single by breaking ties on the basis of secondary criteria, such as simplicity [48] or operational cost [50]

Bayesian approaches explicitly capture uncertainty over the set of competing models by means of a posterior probability distribution. In practice, however, they use the posterior distribution to construct a single model for deployment [e.g., via a majority vote or a randomization procedure as in PAC Bayesian approaches, 18, 34].

These approaches, which are typically designed to improve out-of-sample performance, may not always resolve the irreconcilable differences in predictions between competing models. In some prediction tasks, for examples, tie-breaking between models would not achieve its intended goal of improving out-of-sample performance (e.g. in Figure 1 if we assume that the training data is drawn according to the true underlying distribution).

2 Framework

In this section, we introduce measures to describe the multiplicity of predictions in classification. We restrict our attention to binary classification problems where we want to maximize accuracy. However, our measures can be generalized to settings where we work with three or more outcomes, where we care about different performance metrics (e.g., AUC, TPR), or where models must obey constraints [e.g., fairness constraints 10, 13, 57].

2.1 Preliminaries

We consider a standard binary classification problem. We start with a training dataset of n examples $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where each *example* consists of a vector of *features* $\mathbf{x}_i = (1, x_{i1}, \dots, x_{id}) \in \mathbb{R}^{d+1}$ and a *label* $y_i \in \{-1, +1\}$.

Our goal is to train a classifier $h : \mathbb{R}^{d+1} \rightarrow \{-1, +1\}$ from a hypothesis class $\mathcal{H}$ that optimizes true risk (i.e., test error). To this end, we minimize empirical risk (i.e., training error):

$$\hat{R}(h) := \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq y_i].$$

Formally, a prediction problem consists of a dataset, a hypothesis class, and a specific empirical risk minimization (ERM) problem. We denote an optimal classifier as:

$$\hat{h} \in \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}(h).$$

Note that there may exist multiple optimal classifiers (see e.g., Figure 1).

We assume that we deploy a baseline classifier $\hat{h}_0 = \hat{h}$ that optimizes performance on training data.¹ This practice implicitly assumes that we expect the performance of $\hat{h}$ on the training data to generalize, which is a reasonable assumption in the applications that we consider as we fit classifiers from a simple hypothesis class on datasets where $n > d$ (see also our empirical results in Table 1 in Section 4).

The Set of Competing Models. We measure the multiplicity of predictions among the set of “good” models – i.e., classifiers in $\mathcal{H}$ whose performance differs from the performance of a baseline classifier by a fixed amount. Given a baseline classifier $\hat{h}$, we refer to the set of classifiers whose performance differs by at most ϵ as the ϵ -level set.

DEFINITION 1 (ϵ -LEVEL SET). *Given any classifier h and a hypothesis class $\mathcal{H}$, the ϵ -level set around h is the set of all models $h' \in \mathcal{H}$ that make at most $\hat{R}(h) + \epsilon$ mistakes over the training data:*

$$S_\epsilon(h) := \{h' \in \mathcal{H} : |\hat{R}(h') - \hat{R}(h)| \leq \epsilon\}.$$

We refer to models in the ϵ -level set as *competing models*. Since we measure performance in terms of the error rate, we refer to ϵ as the *error tolerance*.

DEFINITION 2 (MULTIPLICITY). *Given a baseline classifier $\hat{h}_0$ and an error tolerance ϵ , a prediction problem exhibits multiplicity if the ϵ -level set around $\hat{h}_0$ contains at least two models. Formally, $|S_\epsilon(\hat{h}_0)| \geq 2$.*

Prediction problems can admit competing models that assign conflicting predictions yet attain the best possible performance – i.e., predictive multiplicity may arise over an ϵ -level set where $\epsilon = 0$. In practice, however, there is good reason to measure multiplicity over an ϵ -level set where $\epsilon > 0$. This is because there is some chance that competing models that attain near-optimal performance on the training set may outperform the optimal model in deployment (see e.g., Section 4 for empirical evidence). In such settings, it would not be defensible to rule out near-optimal models on the basis of small differences in training error.

¹Deploying a model that optimizes performance on training data reflects best practices, even in settings where we use a validation dataset to tune parameters or a test dataset to estimate out-of-sample performance[see e.g., 9, for a discussion]. For example, in a setting where we control overfitting by tuning hyperparameters, we would first find a set of hyperparameters that optimizes an estimate of error in deployment (e.g., mean 5-CV error). We would then train a model that optimizes performance using all of the available data.

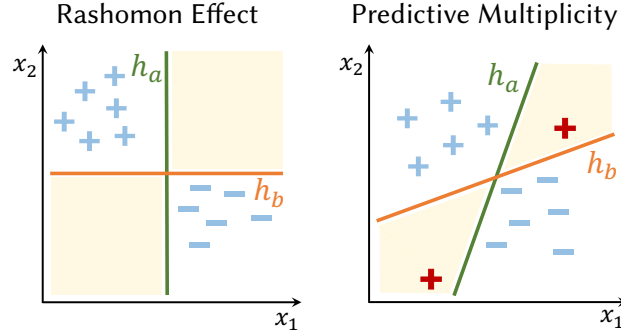


Fig. 2. The “Multiplicity of Good Models” does not necessarily imply a “Multiplicity in Prediction.” We compare classification problems where we can find two classifiers that optimize accuracy: h_a and h_b . We highlight regions of conflict in yellow, and points that are assigned conflicted predictions in red. On the left, h_a and h_b assign the same predictions on the training data (but produce conflicting explanations of the importance of x_1 vs. x_2 , as per the Rashomon effect). On the right, h_a and h_b assign conflicting predictions on the training data (i.e., predictive multiplicity). A prediction problem can exhibit both effects, but predictive multiplicity requires that there is concrete evidence of irreconcilable differences between the predictions of good models.

In practice, one should therefore evaluate multiplicity among a set of models that plausibly contains a model that performs the best in deployment. That is, we should set ϵ so that the ϵ -level set will, with high probability, contain the model that optimizes performance in deployment.²

2.2 Predictive Multiplicity

A prediction problem exhibits *predictive multiplicity* if competing models that perform almost equally well assign conflicting predictions over the training data.

DEFINITION 3 (PREDICTIVE MULTIPLICITY). *Given a baseline classifier $\hat{h}_0$ and an error tolerance ϵ , a prediction problem exhibits predictive multiplicity over the ϵ -level set $S_\epsilon(\hat{h}_0)$ if there exists a model $h \in S_\epsilon(\hat{h}_0)$ such that $h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)$ for some $\mathbf{x}_i$ in the training dataset.*

Predictive multiplicity ignores cases where “good models” are “good” because they predict in the same way. The fact that competing models assign conflicting predictions means that model selection will involve arbitrating irreconcilable predictions.

2.3 Measures of Predictive Multiplicity

Discrepancy. Given a baseline classifier $\hat{h}_0$, we measure the severity of predictive multiplicity by counting the number of conflicting predictions between $\hat{h}_0$ and a model in the ϵ -level set.

²The value of ϵ can be set systematically by generalization bounds, or by computing confidence intervals of out-of-sample performance (e.g., via bootstrapping or cross-validation).

DEFINITION 4 (DISCREPANCY). *The discrepancy over the ϵ -level set $S_\epsilon(\hat{h}_0)$ is the maximum number of conflicting predictions between the baseline classifier $\hat{h}_0$ and any other competing classifier $g_\epsilon \in S_\epsilon(\hat{h}_0)$:*

$$\delta_\epsilon(h) := \max_{h' \in S_\epsilon(\hat{h}_0)} \frac{1}{n} \sum_{i=1}^n 1[h'(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)].$$

If discrepancy is small, it means that competing models must output similar predictions. This reflects the common intuition that models which perform similarly output similar predictions. If discrepancy is large, this means there exists a competing model that makes very different predictions than the baseline classifier. This reflects inherent trade-offs between predictions, possibly resulting from noisy data or structure in the data the model class cannot fit.

Ambiguity. If there exists a classifier in the ϵ -level set making a different prediction on $\mathbf{x}_i$, then the prediction $\hat{h}_0(\mathbf{x}_i)$ may change due to model choice. In this case, we say that the prediction on $\mathbf{x}_i$ is *ambiguous*. Ambiguity has important implications. For example, it may provide grounds for a person to contest their prediction. In addition, it would calibrate trust in prediction-based explanations – i.e., given that there exists a competing model h such that $h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)$, one is less inclined to rationalizing the explanation produced by the deployed model.

We define the *ambiguity* of a model as the number of instances that exhibit individual ambiguity. While discrepancy reflects the maximum number of predictions that can change between the baseline model and a single model in the ϵ -level set, ambiguity measures how many individual predictions are determined by the choice of one model between a set of good models.

DEFINITION 5 (AMBIGUITY). *The ambiguity of a baseline classifier $\hat{h}_0$ over the ϵ -level set $S_\epsilon(\hat{h}_0)$ is the number of points in a training dataset $\mathbf{x}_i \in X$ that can be assigned a conflicting prediction by a competing classifier $g_i \in S_\epsilon(\hat{h}_0)$:*

$$\alpha_\epsilon(\hat{h}_0) := \#\{\mathbf{x}_i \in X : \exists h' \in S_\epsilon(\hat{h}_0) \text{ s.t. } h'(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)\}.$$

Canonical Measures. Ambiguity and discrepancy capture salient aspects of predictive multiplicity for a given prediction problem. Considering the fact that these measures depend on our choice of baseline model, we view the ambiguity and discrepancy of an optimal model as the canonical measures of predictive multiplicity for a given prediction problem, and denote these quantities as α_ϵ and δ_ϵ , respectively. We observe that $\delta_\epsilon \leq \alpha_\epsilon$, since the predictions flipped by a single model are a subset of those flipped by any model. Since different subsets of predictions can be flipped by different models within a level set, there may be prediction problems where ambiguity far exceeds the discrepancy. This would reflect an unintuitive case where all good models make similar decisions, but many points are given conflicting predictions by good models. In other cases, ambiguity and discrepancy are similar to one another. This reflects that a single model is capable of expressing most of the predictive multiplicity.

2.4 Formal Guarantees

We end this section with simple bounds that illustrate the relationship between accuracy and predictive multiplicity. In Proposition 2.1, we bound the number of prediction profiles produced by models in the ϵ -level. In Proposition 2.2, we bound the number of conflicts between the optimal model and any model in the ϵ -level set. We include proofs in Appendix A.

PROPOSITION 2.1 (BOUND ON THE NUMBER OF GOOD MODELS). *Given any value of $\epsilon \in \{0, \dots, n\}$, the number of distinct classifiers in an ϵ -level set of a baseline classifier $\hat{h}_0$ is bounded by*

$$|S_\epsilon(\hat{h}_0)| \leq \sum_{i=\epsilon^{\text{lb}}}^{\epsilon^{\text{ub}}} \binom{n}{\hat{R}(\hat{h}_0) + i}$$

Here, $\epsilon^{\text{lb}} = \max(-\epsilon, -\hat{R}(\hat{h}))$ and $\epsilon^{\text{ub}} = \min(\epsilon, n - \hat{R}(\hat{h}))$ represent lower and upper bounds for error tolerance given a baseline classifier that makes $\hat{R}(\hat{h}_0) \in [0, n]$ mistakes.

PROPOSITION 2.2 (BOUND ON DISCREPANCY). *The discrepancy between $\hat{h}_0$ and any competing classifier in the ϵ -level set $h \in S_\epsilon(\hat{h}_0)$ obeys:*

$$\delta_\epsilon \leq 2\hat{R}(\hat{h}_0) + \epsilon.$$

Propositions 2.1 and 2.2 demonstrate how the severity of predictive multiplicity depends on the accuracy of a baseline model. These bounds illustrate the “pinching” effect of accuracy on multiplicity. As the number of mistakes decreases, the maximum number of possible trade-offs between conflicting points also decreases. In other words, there is more “room” for predictive multiplicity in settings where models perform poorly.

3 Methodology

In this section, we present integer programming tools to compute discrepancy and ambiguity for linear classification problems.

3.1 Preliminaries

Overview. We compute discrepancy and ambiguity by training linear classifiers from the ϵ -level set. We train three kinds of linear classifiers:

1. $\hat{h}$: a baseline classifier that minimizes the training error. While we can compute ambiguity and discrepancy with respect to any baseline classifier, computing them with respect to $\hat{h}$ yields meaningful measures of multiplicity since it reflects the best possible model that can be produced for a given prediction problem.³
2. g_ϵ : a classifier in the ϵ -level set that maximizes the number of conflicting predictions. This classifier will be used to compute discrepancy.
3. g_i : a *pathological classifier* that minimizes training error subject to the constraint that it must flip the prediction of $\hat{h}$ on $\mathbf{x}_i$. We will use g_i to determine the individual ambiguity of i . By training g_i for all $i \in \{1, \dots, n\}$, we will be able to compute ambiguity.

Integer Programming. We train each classifier by solving an ERM problem with discrete constraints. We express each problem as a *mixed integer program* (MIP) that we solve with a MIP solver such as CPLEX [23] or CBC [16]. MIP solvers return the global optimum of non-convex optimization problem

³Computing these measures with respect to the best baseline, we rule out the possibility that multiplicity arises due to a “suboptimal” model. Thus, the only way to avoid multiplicity is to change the prediction problem (e.g., by changing the dataset, incorporating additional requirements, or choosing a different hypothesis class).

using an exhaustive search process [e.g., branch-and-bound 56]. In our setting, solving a MIP produces: (i) an upper bound on the objective value; (ii) a lower bound on the objective value; and (iii) the coefficients of a linear classifier that achieve the upper bound. When the upper bound (i) and the lower bound (ii) match, the solution is said to be *certifiably optimal*.

Path Algorithms. We will present *path algorithms* to efficiently compute our measures for all possible sets of good models – i.e., ϵ -level sets for all possible values of the error tolerance ϵ . [see e.g., algorithms to train machine learning models over the full regularization path such as glmnet 17]. Even as there are systematic ways to set the error tolerance, path algorithms are useful as they avoid the need to set ϵ *a priori* and clearly show how the measures of interest change with respect to the choice of ϵ .

3.2 Training the Baseline Classifier

We train a classifier that minimizes the training error by solving an optimization problem of the form:

$$\min_{h \in \mathcal{H}} \sum_{i=1}^n \mathbb{1}[h(\mathbf{x}_i) \neq y_i] \quad (1)$$

We fit a linear classifier $h(\mathbf{x}_i) = \text{sign}(\langle \mathbf{w}, \mathbf{x}_i \rangle)$ that optimizes the 0-1 loss by solving the MIP formulation:

$$\begin{aligned} \min \quad & \sum_{i=1}^n l_i \\ \text{s.t.} \quad & M_i l_i \geq y_i \left(\gamma - \sum_{j=0}^d w_j x_{ij} \right) \quad i = 1, \dots, n \quad (2a) \\ & 1 = l_i + l_{i'} \quad (i, i') \in K \quad (2b) \\ & w_j = w_j^+ + w_j^- \quad j = 0, \dots, d \quad (2c) \\ & 1 = \sum_{j=0}^d (w_j^+ - w_j^-) \quad (2d) \\ & l_i \in \{0, 1\} \quad i = 1, \dots, n \\ & w_j \in [-1, 1] \quad j = 0, \dots, d \\ & w_j^+ \in [0, 1] \quad j = 0, \dots, d \\ & w_j^- \in [-1, 0] \quad j = 0, \dots, d \end{aligned}$$

Here, constraints (2a) set the mistake indicators $l_i \leftarrow \mathbb{1}[h(\mathbf{x}_i) \neq y_i]$. These constraints depend on: (i) a margin parameter $\gamma > 0$, which should be set to a small positive number (e.g., $\gamma = 10^{-4}$); and (ii) the “Big-M” parameters M_i which can be set as $M_i = \gamma + \max_{\mathbf{x}_i \in X} \|\mathbf{x}_i\|_\infty$ since we have fixed $\|\mathbf{w}\|_1 = 1$ in constraint (2d). Constraint (2b) produces an improved lower bound by encoding the necessary condition that any classifier must make exactly one mistake between any two points $(i, i') \in K$ with identical features $\mathbf{x}_i = \mathbf{x}_{i'}$ and conflicting labels. Here, $K = \{(i, i') : \mathbf{x}_i = \mathbf{x}_{i'}, y_i = +1, y_{i'} = -1\}$ is the set of points with conflicting labels.

3.3 Computing Discrepancy

Given a training dataset, a baseline classifier $\hat{h}_0$, and an error tolerance ϵ , we compute the discrepancy among all models within the ϵ -level set of $\hat{h}_0$ by solving the optimization problem:

$$\begin{aligned} \min_{h \in \mathcal{H}} \quad & \sum_{i=1}^n \mathbb{1} \left[h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i) \right] \\ \text{s.t.} \quad & \hat{R}(h) \leq \hat{R}(\hat{h}_0) + \epsilon \end{aligned} \quad (3)$$

The optimization problem in (3) minimizes the agreement between g_ϵ and $\hat{h}_0$. Given that this is equivalent to maximizing discrepancy, the solution to this optimization problem is a classifier in the ϵ -level set that maximizes discrepancy with respect to the baseline classifier, which we denote as g_ϵ .

We train a linear classifier g_ϵ that maximizes discrepancy with the baseline classifier $\hat{h}_0$ by solving the MIP formulation:

$$\begin{aligned} \max \quad & \sum_{i=1}^n \delta_i \\ \text{s.t.} \quad & M_i l_i \geq \hat{y}_i \left(\gamma - \sum_{j=0}^d w_j x_{ij} \right) \quad i = 1, \dots, n \end{aligned} \quad (4a)$$

$$M_i (1 - \delta_i) \geq \gamma + \hat{y}_i \sum_{j=0}^d w_j x_{ij} \quad i = 1, \dots, n \quad (4b)$$

$$\epsilon \geq \sum_{i=1}^n \delta_i (1 - 2l_i) \quad (4c)$$

$$1 = l_i + l_{i'} \quad (i, i') \in K \quad (4d)$$

$$w_j = w_j^+ + w_j^- \quad j = 0, \dots, d \quad (4e)$$

$$1 = \sum_{j=0}^d (w_j^+ - w_j^-) \quad (4f)$$

$$\delta_i \in \{0, 1\} \quad i = 1, \dots, n$$

$$l_i \in \{0, 1\} \quad i = 1, \dots, n$$

$$w_j \in [-1, 1] \quad j = 0, \dots, d$$

$$w_j^+ \in [0, 1] \quad j = 0, \dots, d$$

$$w_j^- \in [-1, 0] \quad j = 0, \dots, d$$

Here, the objective maximizes the expression $\sum_{i=1}^n \delta_i$, where δ_i is an indicator variable that is set as $\mathbb{1} \left[h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i) \right]$. These variables are set via the “Big-M” constraints (4a) and (4b), which have the same form as constraints (2a). The $\hat{y}_i$ represent the prediction of the baseline classifier on point $\mathbf{x}_i$. Constraint (4c) enforces that any classifier must belong to the ϵ -level set, i.e. that $|\hat{R}(h) - \hat{R}(\hat{h}_0)| \leq \epsilon$. All other constraints and variables are identical to those in MIP (2).

Given a baseline classifier $\hat{h}$ that is provably optimal, we reduce computation by adding the constraint $\sum_{i=1}^n \delta_i (1 - 2l_i) \geq 0$. This constraint reflects the fact that we cannot increase the error rate by using an optimal baseline.

For the discrepancy MIP, the objective value is the maximum discrepancy found by the solver in the ϵ -level set. The upper bound is the maximum possible discrepancy. When the objective value equals the upper bound, the solution provably maximizes the number of conflicts with the baseline classifier.

Path Algorithm. In Algorithm 1, we present a procedure to compute discrepancy for all possible values of ϵ . The procedure solves the MIP (4) for all values of the error tolerance from ϵ^{lb} to ϵ^{ub} . At each iteration, it uses solutions obtained from earlier iterations to initialize the current MIP. This resulting process is many times faster than solving the MIP separately for various error tolerances.

Algorithm 1 Compute Discrepancy for All Possible Values of ϵ

Input

$\hat{h}_0$ baseline classifier
 $\epsilon^{\text{lb}}, \epsilon^{\text{ub}}$ bounds on the error tolerance ϵ (see Section 2)

Initialize

$S \leftarrow \emptyset$ pool of classifiers for initialization
 $\mathcal{P} \leftarrow \emptyset$ value of discrepancy for each ϵ

for $\epsilon \in \{\epsilon^{\text{lb}}, \epsilon^{\text{lb}} + 1, \dots, \epsilon^{\text{ub}}\}$ **do**
 MIP $\leftarrow$ Instance of MIP (4) for error tolerance of ϵ
 Set lower bound of MIP to discrepancy in $\mathcal{P}$
 $g_\epsilon \leftarrow$ solution to MIP
 $\delta_\epsilon \leftarrow$ number of conflicts between g_ϵ and $\hat{h}_0$
 $S \leftarrow S \cup (g_\epsilon)$
 $\mathcal{P} \leftarrow \mathcal{P} \cup (\epsilon, \delta_\epsilon)$

end for

Output: $\mathcal{P}$ the discrepancy for each ϵ considered

3.4 Computing Ambiguity

Given a training dataset and a baseline classifier $\hat{h}_0$, we ambiguity for all possible values of ϵ by fitting a *pathological classifier* g_i for each point $i \in \{1, \dots, n\}$. The pathological classifier for point i g_i is the most accurate linear classifier that outputs a conflicting prediction on point i . We can train g_i by solving an optimization problem of the form:

$$\begin{aligned} \min_{h \in \mathcal{H}} \quad & \sum_{k=1}^n \mathbb{1}[h(\mathbf{x}_k) \neq y_k] \\ \text{s.t.} \quad & h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i) \end{aligned} \tag{5}$$

Here, $h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)$ forces any feasible classifier h to assign a conflicting prediction to point $\mathbf{x}_i$. Seeing how this optimization problem is identical to the one in (1), we train this model using a variant of the MIP in (2) that has includes the following constraint to ensure $h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)$:

$$\gamma \leq -\hat{h}_0(\mathbf{x}_i) \sum_{j=0}^d w_j x_{ij}. \tag{6}$$

The solution to this MIP returns the coefficients and training error of the classifier g_i . With these quantities in hand, we can determine if point i exhibits individual ambiguity over the set of ϵ -good models by checking if the following inequality holds:

$$\hat{R}(g_i) \leq \hat{R}(\hat{h}_0) + \epsilon.$$

Given a set of pathological classifiers g_i for each point $i \in \{1, \dots, n\}$, we can compute ambiguity over the ϵ -level set by counting the number of pathological classifiers whose performance is within ϵ of the baseline model. That is:

$$\alpha_\epsilon = \#\{\mathbf{x}_i \in X : \hat{R}(g_i) \leq \hat{R}(\hat{h}_0) + \epsilon\}$$

Algorithm 2 Computing Ambiguity for all Possible Values of ϵ

Input

$\hat{h}_0$ baseline classifier
 $\epsilon^{\text{lb}}, \epsilon^{\text{ub}}$ bounds on the error tolerance ϵ (see Section 2)

Initialize

$\mathcal{S} \leftarrow \emptyset$ pool of classifiers for initialization
 $\Delta \leftarrow \emptyset$ training error increase to change each prediction
 $\mathcal{P} \leftarrow \emptyset$ value of ambiguity for each ϵ

for $i \in \{1, 2, \dots, n\}$ **do**

 MIP $\leftarrow$ Instance of MIP (6) for point $\mathbf{x}_i$
 Initialize MIP to best solution in $\mathcal{S}$ satisfying Equation (6)
 $g_i \leftarrow$ solution to MIP
 $\Delta_i \leftarrow$ increase in train error of g_i versus $\hat{h}_0$
 $\mathcal{S} \leftarrow \mathcal{S} \cup (g_i)$
 $\Delta \leftarrow \Delta \cup (\Delta_i)$

end for

for $\epsilon \in \{\epsilon^{\text{lb}}, \epsilon^{\text{lb}} + 1, \dots, \epsilon^{\text{ub}}\}$ **do**

$\alpha_\epsilon \leftarrow$ number of Δ_i less than ϵ
 $\mathcal{P} \leftarrow \mathcal{P} \cup \alpha_\epsilon$

end for

Output: $\mathcal{P}$

the ambiguity for each ϵ considered

Our procedure involves substantial computation as we must train a pathological classifier for each $\mathbf{x}_i$. In our implementation, we reduce the computation in the following ways:

- We collect the solutions from previous MIPs, and initialize each MIP with the best solution that obeys $h(\mathbf{x}_i) \neq \hat{h}_0(\mathbf{x}_i)$.
- We initialize the lower bound of the MIP that we solve to train g_i to the lower bound produced by the baseline MIP.
- We add a constraint to fix $l_i = \mathbb{1} \left[\hat{h}_0(\mathbf{x}_i) = y_i \right]$ in MIP (6). Since $\hat{h}_0$ and g_i must output conflicting predictions, it follows that exactly one of these classifiers must make a mistake on $\mathbf{x}_i$.

3.5 Discussion

Our use of integer programming is motivated by the fact that it can directly optimize and constrain the quantities that we care about. By using integer programming, we can train classifiers in a way that considers *all* possible models in an ϵ -level set. In contrast, popular approaches to reduce computation (e.g., solving a ERM problem with a convex surrogate loss function such as the hinge loss) may overlook competing models in the ϵ -level set, and thus return unreliable estimates.

Although integer programming is computationally intractable, modern solvers can solve ERM problems for many real-world datasets [see e.g., 6, 40, 51–53]. Using a solver also has the benefit that it pairs each solution with a bound on optimal. Thus, when integer programs do not produce a certifiably optimal solution, they return a feasible solution along with bounds on the objective value that can be used to bound our measures of predictive multiplicity. For example, in cases where the discrepancy MIP does not solve to optimality, the solver would return the best possible solution along with a lower bound on the minimum agreement (i.e., an upper bound on the discrepancy).

Extensions. Our MIP formulations can be modified to optimize other objectives such as measures of class-based accuracy and to obey constraints on model form or model predictions. For instance, we can measure the ambiguity and discrepancy of linear classifiers that obey exact constraints on group fairness. Likewise, we can handle rule-based models that can be expressed as special kinds of linear classifiers, such as decision sets [30], decision lists [3], conjunctions and disjunctions [55].

4 Experiments

In this section, we apply our tools to evaluate predictive multiplicity in recidivism prediction problems. We have four goals: (i) to measure the incidence of predictive multiplicity in real-world classification problems; (ii) to show how our tools can inform stakeholders; (iii) to discuss the implications of predictive multiplicity with respect to transparency and accountability; (iv) to show that we also measure predictive multiplicity using existing tools (albeit imperfectly). We include the software, scripts, and datasets to generate these results at <https://github.com/charliemarx/pmtools>. We consider recidivism prediction because it is a domain where predictive multiplicity has ethical implications, and because it may present overlooked critique of existing practices [see e.g., 5, 20, 32, for existing critiques].

4.1 Setup

Datasets. We define a set of recidivism prediction problems from three studies of recidivism in the United States:

`compas`: ProPublica COMPAS repository [4].

`pretrial`: [Felony Defendants in Large Urban Counties](#) [43] [used by e.g., 19, 25];

`recidivism`: [Recidivism of Prisoners Released in 1994](#) [42] [used by e.g., 1, 58];

We use the data from each study to define binary classification problems by defining an outcome variable (e.g., a proxy variable such as arrest, arrest for violent crime) and a set of features for prisoners released in a specific state (e.g., NY, CA). Following standard practices in the development of recidivism

Dataset	Outcome Variable	n	d	Error of Baseline Model	
				Training Set	Test Set
compas_arrest	rearrest for any crime	5,380	18	32.7%	33.4%
compas_violent	rearrest for violent crime	8,768	18	37.7%	37.9%
pretrial_CA_arrest	failure to appear	9,926	22	34.1%	34.4%
pretrial_CA_fta	failure to appear	8,738	22	36.3%	36.3%
recidivism_CA_arrest	rearrest for any offense	114,522	20	34.4%	34.2%
recidivism_CA_drug	rearrest for a drug-related offense	96,664	20	36.3%	36.2%
recidivism_NY_arrest	rearrest for any offense	31,624	20	31.0%	31.8%
recidivism_NY_drug	rearrest for a drug-related offense	27,526	20	32.5%	33.6%

Table 1. Overview of recidivism prediction problems. For each dataset, we train a baseline linear classifier that we would deploy model by minimizing the 0-1 loss on the training data. As shown, the models generalize as training error is close to test error. This is expected given that we fit models from a simple hypothesis class on datasets where $n \gg d$. Here, n is the number of instances and d is the number of features in the dataset.

prediction instruments, we also binarize features and drop instances with missing entries [see e.g., guidelines for the development of an instrument by the Pennsylvania Sentencing Commission 45].

We oversample the minority class to equalize the number of instances with positive and negative labels. Oversampling ensures that we can directly minimize the training error without producing trivial classifiers. This also allows us to report meaningful measures of the error rate rather than class-specific error rates. We find that oversampling has a trivial effect on our measures of multiplicity ($< 1\%$).

We present an overview of the number of instances and features for each dataset in Table 1. Each dataset are publicly available. We include a copy of `compas_violent` and `compas_arrest` along with our code. The remaining datasets must be requested from ICPSR due to privacy restrictions.

Measurement with Our Tools. We evaluate the degree of predictive multiplicity for each dataset as follows. We split each dataset into a *training set* composed of 80% of points and a *test set* composed of 20% of points. We use the training set to fit a *baseline classifier* by solving MIP (2). We then measure the discrepancy and ambiguity among the set of good models by solving the MIPs in Sections 3.4 and 3.3 for *all possible values* of the error tolerance ϵ . To measure discrepancy, we solve MIP (4) for all possible values of the error tolerance $\epsilon \in [0, n/2]$. To measure ambiguity, we solve MIP (6) for each distinct point in the training data. We solve all MIPs on a 3.33 GHz CPU with 16 GB RAM. We allocate at most 6 hours to train the baseline model, 6 hours to train all discrepancy models, and 6 hours to train the individual ambiguity models.

Measurement using Standard Methods. We also consider an alternative analysis where we compute our measures of predictive multiplicity over a large pool of linear classifiers. We include this analysis to show that a systematic evaluation of predictive multiplicity can still reveal salient information – even without using specialized techniques.

We construct a pool of linear classifiers via penalized logistic regression using the `glmnet` package of [17]. Each model optimizes a specific weighted combination of the ℓ_1 and ℓ_2 penalties. Thus, the

coefficients of each model are the solution to the optimization problem:

$$\min_{\mathbf{w} \in \mathbb{R}^{d+1}} \frac{1}{2n} \sum_{i=1}^n \log(1 + \exp(-\langle \mathbf{w}, y_i \mathbf{x}_i \rangle)) + \lambda \cdot \left(\alpha \|\mathbf{w}\|_1 + (1 - \alpha) \|\mathbf{w}\|_2^2 \right) \quad (7)$$

Here $\alpha \in [0, 1]$ is the elastic-net mixing parameter and $\lambda \geq 0$ is a regularization penalty. We train 1,100 models by choosing the following 1,100 combinations of (α, λ) : 11 values of $\alpha \in \{0.0, 0.1, \dots, 0.9, 1.0\} \times 100$ values of λ (chosen automatically by glmnet for each α). This produces a pool of 1,100 models that includes models produced by: (i) Lasso (ℓ_1 -penalty), which corresponds to when $\alpha = 1.0$; (ii) Ridge (ℓ_2 -penalty), which corresponds to when $\alpha = 0.0$; (iii) standard logistic regression, which corresponds to when $\alpha = 0.0$ and λ is small.

Our baseline model minimizes 5-fold cross-validation error. We compute discrepancy and ambiguity by comparing the baseline model to models in the pool with at most ϵ additional training error.

	Baseline Model	Individual Ambiguity Model	Discrepancy Model
$h(\mathbf{x}_p)$	+1	-1	-1
Error (Train)	32.7%	32.7%	33.6%
Error (Test)	33.4%	33.4%	34.5%
Discrepancy (Train)	0.0%	0.0037%	16.8%
Discrepancy (Test)	0.0%	0.0%	15.1%
Score Function	+ 0.5 age $\leq$ 25	+ 10.3 age $\leq$ 25	+ 7.7 age $\leq$ 25
	+ 0.0 age_25-45	+ 0.0 age_25-45	+ 0.0 age_25-45
	- 16.4 age $\geq$ 46	- 9.9 age $\geq$ 46	- 7.8 age $\geq$ 46
	- 16.3 female	- 9.7 female	- 7.6 female
	- 0.2 n_priors=0	+ 0.0 n_priors=0	- 7.8 n_priors=0
	- 0.1 n_priors $\geq$ 1	+ 0.0 n_priors $\geq$ 1	+ 0.0 n_priors $\geq$ 1
	+ 16.4 n_priors $\geq$ 2	+ 19.8 n_priors $\geq$ 2	+ 7.4 n_priors $\geq$ 2
	+ 16.6 n_priors $\geq$ 5	+ 10.1 n_priors $\geq$ 5	+ 7.8 n_priors $\geq$ 5
	+ 0.0 n_juvenile_misdemeanors=0	+ 0.0 n_juvenile_misdemeanors=0	+ 0.0 n_juvenile_misdemeanors=0
	- 0.1 n_juvenile_misdemeanors $\geq$ 1	- 0.1 n_juvenile_misdemeanors $\geq$ 1	+ 0.1 n_juvenile_misdemeanors $\geq$ 1
	+ 0.0 n_juvenile_misdemeanors $\geq$ 2	- 10.1 n_juvenile_misdemeanors $\geq$ 2	- 0.1 n_juvenile_misdemeanors $\geq$ 2
	- 32.6 n_juvenile_misdemeanors $\geq$ 5	- 9.5 n_juvenile_misdemeanors $\geq$ 5	- 15.2 n_juvenile_misdemeanors $\geq$ 5
	+ 0.0 n_juvenile_felonies=0	- 9.9 n_juvenile_felonies=0	+ 7.7 n_juvenile_felonies=0
	- 0.2 n_juvenile_felonies $\geq$ 1	- 10.1 n_juvenile_felonies $\geq$ 1	+ 0.0 n_juvenile_felonies $\geq$ 1
	+ 0.3 n_juvenile_felonies $\geq$ 2	+ 0.3 n_juvenile_felonies $\geq$ 2	+ 15.4 n_juvenile_felonies $\geq$ 2
	+ 0.0 n_juvenile_felonies $\geq$ 5	+ 0.0 n_juvenile_felonies $\geq$ 5	+ 0.0 n_juvenile_felonies $\geq$ 5
	- 0.2 charge_degree=M	- 0.2 charge_degree=M	- 7.5 charge_degree=M
	+ 0.0	+ 0.0	- 0.1

Table 2. Competing linear classifiers for `compas_arrest`, and their predictions on a person with features $\mathbf{x}_p$. We show the baseline model that we would deploy (left), the models trained to check individual ambiguity (middle), and the model trained to measure discrepancy (right). The baseline model predicts $h(\mathbf{x}_p) = +1$ while the other models predict $h(\mathbf{x}_p) = -1$. As shown, there exists at least two competing models that predict that $\mathbf{x}_p$ would not recidivate. Moreover, each model exhibits different coefficients and measures of variable importance. Models in the 1%-level set exhibit an ambiguity of 44%, meaning that 44% of individuals could have been assigned a different prediction by a competing model with comparable error.

4.2 Results

In Figure 3, we present plots for the discrepancy and ambiguity over all possible values of the error tolerance ϵ . We show plots produced using our tools for three datasets, and plots produced using ad hoc analysis for one dataset. We include for plots for all other datasets in Appendix B. In Table 2, we present

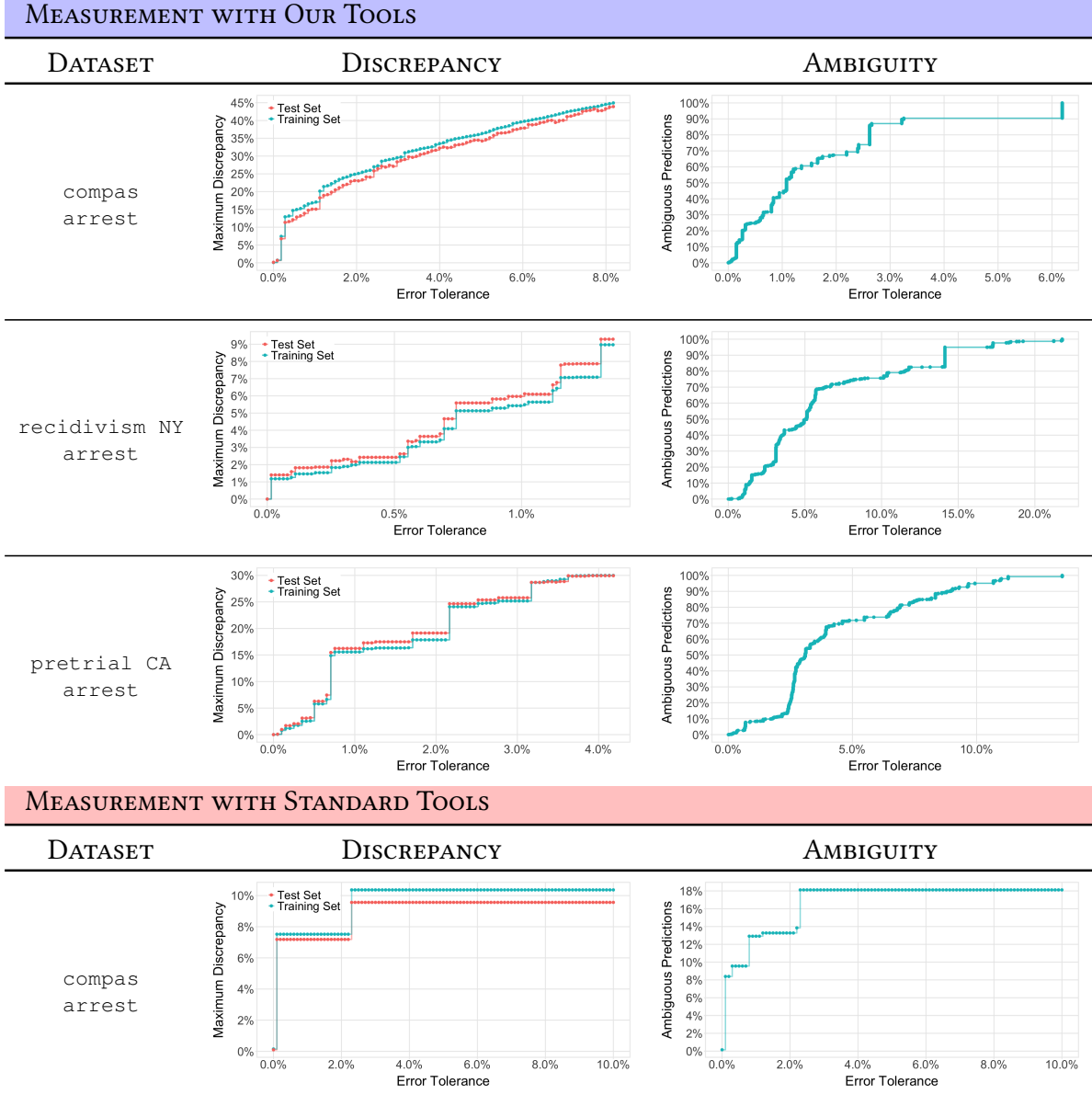


Fig. 3. Overview of predictive multiplicity measures obtained with our tools (top) and using standard tools (bottom). We plot the values of discrepancy (left) and ambiguity (right) over the ϵ -level set (i.e., the set of models whose training error is within ϵ of the optimal training error). Our analysis for `compas_arrest` shows that the set of models for $\epsilon = 1\%$ exhibit a discrepancy of 17%, and an ambiguity of 44%. This means that one can change 17% of predictions by switching the baseline model with a model that is only 1% less accurate. Ambiguity reflects the fact that 44% of predictions are susceptible to model selection within this set. We observe similar values of discrepancy and ambiguity on both the training and test set.

a comparison of the competing classifiers trained on `compas_arrest` in Table 2. In what follows, we discuss these results. Unless otherwise specified, we discuss the values of ambiguity and discrepancy obtained for an error tolerance of $\epsilon = 1\%$, which represents a conservative default.

On the Incidence of Predictive Multiplicity. Our results in Figure 3 shows how predictive multiplicity arises often in real-world prediction tasks. We find that between 4% and 53% of individuals are assigned an ambiguous prediction by models in the 1%-level set. In `compas_arrest`, for example, we observe an ambiguity of 44%. This means that 44% of individuals are assigned conflicting predictions by a competing models. In `compas_violent`, ambiguity over the 1%-level set is 53%, which means that *the majority of predictions are affected by the decision to choose a model within the 1% training error*. Considering discrepancy, we see that `compas_arrest` and `compas_violent` we can find *a single competing model* in the 1%-level set that assigns a conflicting prediction to 17% of individuals. As shown in Figure 3, these measures of multiplicity generalize to the test set.

Our results illustrate why it not be defensible to rule out competing models due to small differences in training performance. Even in settings where we fit the best possible model that optimizes our performance objectives on the training data, this model may not attain the best possible performance in deployment. In `compas_arrest`, for example, we find that the model that optimizes test error is only in the set of good models when $\epsilon \geq 1\%$.

As shown in Table 1, we can train baseline classifiers that perform well and that generalize for each dataset. This is expected given that we are fitting linear classifiers on large datasets via 0-1 loss minimization. Here, we expect the models generalize since we are fitting classifiers from a simple hypothesis class, using a large dataset, and with an algorithm that is robust to outliers [see 41].

On the Implications of Predictive Multiplicity. Our results illustrate how reporting ambiguity and discrepancy can challenge fundamental tasks in model development and deployment. In `compas_arrest`, for example, our baseline model training error and appears to generalize. In the absence of an analysis of predictive multiplicity, many practitioners would deploy this model. However, our analysis reveals that exists a competing model that has similar train and test performance yet assigns conflicting predictions to 17% of individuals. Thus, these measures support the need for greater scrutiny and support stakeholder involvement in model selection (as per the principles of contestable design).

Large values of ambiguity and discrepancy also lead us to calibrate trust in downstream processes. For example, awareness that 44% of predictions are determined by model selection in the 1%-level set calibrates trust in downstream processes that depend on predicted outcomes. Consider the process of explaining individual predictions. In this case, an ambiguity of 44% of points means we could produce conflicting explanations for individual predictions (e.g., a counterfactual explanation for the deployed model could explain how to change $h(\mathbf{x}_p) = -1$ to $+1$ while a counterfactual explanation for a competing model could explain how to change $h(\mathbf{x}_p) = -1$ to $+1$). While both explanations would reflect how the model operates, awareness of the fact that these explanation would change based on model selection would calibrate trust in explanations and provide a safeguard against unwarranted rationalization.

On the Burden of Multiplicity. We find that the incidence of multiplicity can differ significantly between subgroups. In `compas_violent`, for example, the proportion of individuals whose predictions are ambiguous within the 1% level set is: 72.9% among African-American individuals ($n = 4,042$), 37.2% among Caucasian individuals ($n = 3,923$), 32.5% among Hispanic individuals ($n = 579$), and 34.4%

among Other individuals ($n = 224$). In this case, we find that multiplicity disproportionately affects African-Americans compared to individuals of other ethnic groups. Groups with a larger burden of multiplicity are more vulnerable to model selection, and more likely to be affected by the ignorance of competing models.

On Model Selection. When presented with a set of near equivalent models, a natural solution is to choose among them to optimize other goals. We support this practice in cases where secondary objectives are justified (see Section 5 for a discussion). However, tie-breaking does not always yield a unique model. For the `compas_arrest` dataset, for example, we find 102 models (i.e. distinct predictions) that differ from the optimal model by less than 1% train error and 1% disparity between African-American individuals and individuals of other ethnicities. Thus, there are cases where secondary objectives fail to sufficiently distinguish between good models.

On Measurement via Ad Hoc Analysis. Our results show that predictive multiplicity can arise even in settings where we fit classifiers by optimizing an objective function that is guaranteed to produce a unique optimum.⁴ These findings illustrate how reporting predictive multiplicity can reveal useful information even without specialized tools. In `compas_arrest`, for example, an ad hoc analysis reveals that models achieve an ambiguity of 10% and a discrepancy of 7% – i.e., 10% of predictions are assigned conflicting predictions by any competing model, and at most 7% can be flipped by a single competing model. These estimates are far less than those produced using our tools (44% and 17% respectively) because we are working with an incomplete set of “good” models.⁵ These results show that ad hoc measure can capture the presence of predictive multiplicity, but should not be used to certify the absence of multiplicity.

5 Concluding Remarks

Predictive multiplicity can arise in many real-world problems, and has important implications for how we build and interact with predictive models. In applications such as recidivism prediction and credit scoring, predictive multiplicity implies irreconcilable differences in predictions. Thus, when a prediction problem exhibits predictive multiplicity, one must carefully justify the decision to choose one model over other models that perform equally well. If there exists different models that produce conflicting predictions on individuals, without a justification to support one model over another, then it may be a better choice to forgo deployment entirely, or to assign favorable predictions to individuals who receive ambiguous predictions.

Even as there exists several techniques to choose between competing models (see Section 1.1), we do not advocate a general prescription to “resolve” predictive multiplicity. Instead, we argue that we should measure and report multiplicity in the same way that we measure and report test error [22, 37, 46, 47]. In this way, problems that exhibit predictive multiplicity can be handled on a case-by-case basis, and in a way that allows for input from stakeholders [as per the principles of contestable design; see e.g., 21, 28]. Our prescription mirrors Breiman’s recommendations against the use of “automatic methods for feature selection.”

⁴The objective is strongly convex since all instances, even Lasso, include a tiny ℓ_2 penalty.

⁵We do not consider all linear classifiers in the 1%-level set; only models produced by varying ℓ_1 and ℓ_2 penalties in Eq. (7)

Prediction problems can exhibit predictive multiplicity due to a host of reasons, including the use of feature selection, training data with missing features or latent groups, or a misspecified model class. In turn, evaluating predictive multiplicity may guide us in choosing a suitable solution. For example, in Figure 1, all models perform equally well and tie-breaking between models on the basis of out-of-sample accuracy would not achieve the goal of improving out-of-sample accuracy. Likewise, evaluating predictive multiplicity requires a systematic way to search for and evaluate alternative models, which can lead to improvements in the model in deployment, or provide safeguards against a malicious practitioner.

One approach to handle predictive multiplicity is to circumvent difficult choices between competing models by adapting model development to output a unique solution. For example, by incorporating unnecessary constraints to the problem, or by choosing an algorithm that is guaranteed to produce a single model. If such changes are only motivated by the need to produce a unique model, rather than to achieve other desirable objectives, then this would “automate” model selection in an unwarranted way.

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References

- [1] Philip Adler, Casey Falk, Sorelle A Friedler, Tionney Nix, Gabriel Rybeck, Carlos Scheidegger, Brandon Smith, and Suresh Venkatasubramanian. 2018. Auditing black-box models for indirect influence. *Knowledge and Information Systems* 54, 1 (2018), 95–122.
- [2] Ulrich Aivodji, Hiromi Arai, Olivier Fortineau, Sébastien Gambs, Satoshi Hara, and Alain Tapp. 2019. Fairwashing: the risk of rationalization. *arXiv preprint arXiv:1901.09749* (2019).
- [3] Elaine Angelino, Nicholas Larus-Stone, Daniel Alabi, Margo Seltzer, and Cynthia Rudin. 2017. Learning certifiably optimal rule lists for categorical data. *The Journal of Machine Learning Research* 18, 1 (2017), 8753–8830.
- [4] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner. 2016. Machine bias. *ProPublica*, May 23 (2016), 2016.
- [5] Chelsea Barabas, Karthik Dinakar, Joichi Ito, Madars Virza, and Jonathan Zittrain. 2017. Interventions over predictions: Reframing the ethical debate for actuarial risk assessment. *arXiv preprint arXiv:1712.08238* (2017).
- [6] Pietro Belotti, Pierre Bonami, Matteo Fischetti, Andrea Lodi, Michele Monaci, Amaya Nogales-Gómez, and Domenico Salvagnin. 2016. On handling indicator constraints in mixed integer programming. *Computational Optimization and Applications* 65, 3 (2016), 545–566.
- [7] Reuben Binns, Max Van Kleek, Michael Veale, Ulrik Lyngs, Jun Zhao, and Nigel Shadbolt. 2018. ‘It’s Reducing a Human Being to a Percentage’: Perceptions of Justice in Algorithmic Decisions. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 377.
- [8] Leo Breiman. 2001. Statistical modeling: The two cultures. *Statistical science* 16, 3 (2001), 199–231.
- [9] Gavin C Cawley and Nicola LC Talbot. 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research* 11, Jul (2010), 2079–2107.
- [10] L Elisa Celis, Lingxiao Huang, Vijay Keswani, and Nisheeth K Vishnoi. 2019. Classification with fairness constraints: A meta-algorithm with provable guarantees. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, 319–328.
- [11] Alexandra Chouldechova. 2017. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big data* 5, 2 (2017), 153–163.
- [12] Sam Corbett-Davies, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. 2017. Algorithmic decision making and the cost of fairness. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 797–806.
- [13] Andrew Cotter, Heinrich Jiang, Serena Wang, Taman Narayan, Seungil You, Karthik Sridharan, and Maya R Gupta. 2019. Optimization with Non-Differentiable Constraints with Applications to Fairness, Recall, Churn, and Other Goals. (2019).

- [14] Jiayun Dong and Cynthia Rudin. 2019. Variable Importance Clouds: A Way to Explore Variable Importance for the Set of Good Models. *arXiv preprint arXiv:1901.03209* (2019).
- [15] Aaron Fisher, Cynthia Rudin, and Francesca Dominici. 2018. All models are wrong but many are useful: Variable importance for black-box, proprietary, or misspecified prediction models, using model class reliance. *arXiv preprint arXiv:1801.01489* (2018).
- [16] John Forest Forrest and Ted Ralphs. 2019. COIN Branch and Cut. <https://projects.coin-or.org/Cbc>.
- [17] Jerome Friedman, Trevor Hastie, and Robert Tibshirani. 2010. Regularization Paths for Generalized Linear Models via Coordinate Descent. *Journal of Statistical Software* 33, 1 (2010), 1–22.
- [18] Pascal Germain, Francis Bach, Alexandre Lacoste, and Simon Lacoste-Julien. 2016. PAC-Bayesian theory meets Bayesian inference. In *Advances in Neural Information Processing Systems*. 1884–1892.
- [19] Ben Green and Yiling Chen. 2019. Disparate interactions: An algorithm-in-the-loop analysis of fairness in risk assessments. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, 90–99.
- [20] Bernard E Harcourt. 2008. *Against prediction: Profiling, policing, and punishing in an actuarial age*. University of Chicago Press.
- [21] Tad Hirsch, Kritzia Merced, Shrikanth Narayanan, Zac E Imel, and David C Atkins. 2017. Designing contestability: Interaction design, machine learning, and mental health. In *Proceedings of the 2017 Conference on Designing Interactive Systems*. ACM, 95–99.
- [22] Sarah Holland, Ahmed Hosny, Sarah Newman, Joshua Joseph, and Kasia Chmielinski. 2018. The dataset nutrition label: A framework to drive higher data quality standards. *arXiv preprint arXiv:1805.03677* (2018).
- [23] IBM ILOG. 2019. CPLEX Optimizer 12.8. <https://www-01.ibm.com/software/commerce/optimization/cplex-optimizer/>.
- [24] Marcia K Johnson and Steven J Sherman. 1990. Constructing and reconstructing the past and the future in the present. (1990).
- [25] Jon Kleinberg, Himabindu Lakkaraju, Jure Leskovec, Jens Ludwig, and Sendhil Mullainathan. 2017. Human decisions and machine predictions. *The quarterly journal of economics* 133, 1 (2017), 237–293.
- [26] Jon Kleinberg, Jens Ludwig, Sendhil Mullainathan, and Ziad Obermeyer. 2015. Prediction policy problems. *American Economic Review* 105, 5 (2015), 491–95.
- [27] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. 2016. Inherent trade-offs in the fair determination of risk scores. *arXiv preprint arXiv:1609.05807* (2016).
- [28] Daniel Kluttz, Nitin Kohli, and Deirdre K Mulligan. 2018. Contestability and Professionals: From Explanations to Engagement with Algorithmic Systems. *Available at SSRN 3311894* (2018).
- [29] Derek J Koehler. 1991. Explanation, imagination, and confidence in judgment. *Psychological bulletin* 110, 3 (1991), 499.
- [30] Himabindu Lakkaraju, Stephen H Bach, and Jure Leskovec. 2016. Interpretable decision sets: A joint framework for description and prediction. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*. ACM, 1675–1684.
- [31] Zachary Lipton, Julian McAuley, and Alexandra Chouldechova. 2018. Does mitigating ML’s impact disparity require treatment disparity?. In *Advances in Neural Information Processing Systems*. 8125–8135.
- [32] Kristian Lum and William Isaac. 2016. To predict and serve? *Significance* 13, 5 (2016), 14–19.
- [33] David Martens and Foster Provost. 2014. Explaining data-driven document classifications. *MIS Quarterly* 38, 1 (2014), 73–100.
- [34] David A McAllester. 1999. Some PAC-Bayesian Theorems. *Machine Learning* 37, 3 (1999), 355–363.
- [35] James W McAllister. 2007. Model selection and the multiplicity of patterns in empirical data. *Philosophy of Science* 74, 5 (2007), 884–894.
- [36] P McCullagh and John A Nelder. 1989. *Generalized Linear Models*. Vol. 37. CRC Press.
- [37] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebu. 2019. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, 220–229.
- [38] Shira Mitchell, Eric Potash, and Solon Barocas. 2018. Prediction-based decisions and fairness: A catalogue of choices, assumptions, and definitions. *arXiv preprint arXiv:1811.07867* (2018).
- [39] Dean Mountain and Cheng Hsiao. 1989. A combined structural and flexible functional approach for modeling energy substitution. *J. Amer. Statist. Assoc.* 84, 405 (1989), 76–87.

- [40] Seth Neel, Aaron Roth, and Zhiwei Steven Wu. 2018. How to use heuristics for differential privacy. *arXiv preprint arXiv:1811.07765* (2018).
- [41] Tan Nguyen and Scott Sanner. 2013. Algorithms for Direct 0-1 Loss Optimization in Binary Classification. 1085–1093.
- [42] United States Department of Justice. Office of Justice Programs. Bureau of Justice Statistics. 2014. Recidivism of Prisoners Released in 1994. (2014). <https://doi.org/10.3886/ICPSR03355.v8>
- [43] United States Department of Justice. Office of Justice Programs. Bureau of Justice Statistics. 2014. State Court Processing Statistics, 1990-2009: Felony Defendants in Large Urban Counties. (2014). <https://doi.org/10.3886/ICPSR02038.v5>
- [44] Samir Passi and Solon Barocas. 2019. Problem formulation and fairness. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, 39–48.
- [45] Pennsylvania Bulletin. 2017. Sentence Risk Assessment Instrument. , 1999 pages.
- [46] Dillon Reisman, Jason Schultz, Kate Crawford, and Meredith Whittaker. 2018. Algorithmic impact assessments: A practical framework for public agency accountability. *AI Now Institute* (2018).
- [47] Pedro Saleiro, Benedict Kuester, Abby Stevens, Ari Anisfeld, Loren Hinkson, Jesse London, and Rayid Ghani. 2018. Aequitas: A Bias and Fairness Audit Toolkit. *arXiv preprint arXiv:1811.05577* (2018).
- [48] Lesia Semenova and Cynthia Rudin. 2019. A study in Rashomon curves and volumes: A new perspective on generalization and model simplicity in machine learning. *arXiv preprint arXiv:1908.01755* (2019).
- [49] Galit Shmueli et al. 2010. To explain or to predict? *Statistical science* 25, 3 (2010), 289–310.
- [50] Theja Tulabandhula and Cynthia Rudin. 2013. Machine learning with operational costs. *The Journal of Machine Learning Research* 14, 1 (2013), 1989–2028.
- [51] Berk Ustun, Yang Liu, and David Parkes. 2019. Fairness without Harm: Decoupled Classifiers with Preference Guarantees. 97 (09–15 Jun 2019), 6373–6382.
- [52] Berk Ustun and Cynthia Rudin. 2015. Supersparse linear integer models for optimized medical scoring systems. *Machine Learning* 102, 3 (Nov 2015), 349–391. <https://doi.org/10.1007/s10994-015-5528-6>
- [53] Berk Ustun and Cynthia Rudin. 2017. Optimized Risk Scores. (2017).
- [54] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2017. Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR. (2017).
- [55] Tong Wang, Cynthia Rudin, Finale Doshi-Velez, Yimin Liu, Erica Klampfl, and Perry MacNeille. 2017. A Bayesian framework for learning rule sets for interpretable classification. *The Journal of Machine Learning Research* 18, 1 (2017), 2357–2393.
- [56] Laurence A Wolsey. 1998. *Integer Programming*. Vol. 42. Wiley New York.
- [57] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P Gummadi. 2019. Fairness Constraints: A Flexible Approach for Fair Classification. *Journal of Machine Learning Research* 20, 75 (2019), 1–42.
- [58] Jiaming Zeng, Berk Ustun, and Cynthia Rudin. 2017. Interpretable classification models for recidivism prediction. *Journal of the Royal Statistical Society: Series A (Statistics in Society)* 180, 3 (2017), 689–722.

A Omitted Proofs

Proof of Proposition 2.1. We proceed by a simple counting argument. First, note that in the context of predictive multiplicity we consider unique models to be models which differ in at least one observed prediction. Now, there are $\binom{n}{\hat{R}(\hat{h}_0)}$ ways to choose $\hat{R}(\hat{h}_0)$ instances on which to make a mistake. Denote the effective lower and upper bounds of the ϵ -level set by $\epsilon^{\text{lb}} = \max(-\epsilon, -\hat{R}(\hat{h}_0))$ and $\epsilon^{\text{ub}} = \min(\epsilon, n - \hat{R}(\hat{h}_0))$ respectively, since the number of mistakes of any model must be in the interval $[0, n]$. Since the ϵ -level set is comprised only of models making a number of mistakes in the interval $[\hat{R}(\hat{h}_0) + \epsilon^{\text{lb}}, \hat{R}(\hat{h}_0) + \epsilon^{\text{ub}}]$, we see that there are at most $\sum_{i=\epsilon^{\text{lb}}}^{\epsilon^{\text{ub}}} \binom{n}{\hat{R}(\hat{h}_0)+i}$ unique prediction vectors a model in the ϵ -level set could generate. Thus, this supplies an upper bound on the number of unique models in the ϵ -level set. Note that the number of these prediction vectors that will be achievable will vary between model classes. $\square$

Proof of Proposition 2.2. We will use the Triangle Inequality to bound the distance between the vector of predictions of the baseline model and the predictions of a competing model in the ϵ -level set. Let $y = \{y_i\}_{i=1}^n$ be the vector of labels, let $\hat{y} = \{\hat{h}_0(\mathbf{x}_i)\}_{i=1}^n$ be the vector of predictions of the baseline model, and let $y' = \{h'(\mathbf{x}_i)\}_{i=1}^n$ be the predictions of a competing model h' in the ϵ -level set. Note that $y, y', \hat{y} \in \{+1, -1\}^n$. Now, we can express the risk of each model, $\hat{R}(\hat{h})$ and $\hat{R}(h')$, as well as the discrepancy between the two models, denoted $\delta(\hat{h}, h')$, in terms of these three vectors by

$$\begin{aligned}\hat{R}(\hat{h}) &= \frac{1}{4} \|y - \hat{y}\| \\ \hat{R}(h') &= \frac{1}{4} \|y - y'\| \\ \delta(\hat{h}, h') &= \frac{1}{4} \|y' - \hat{y}\|\end{aligned}$$

Next, consider the triangle formed in $\mathbb{R}^n$ by the points y, y' and $\hat{y}$. This triangle has side lengths $\|y - \hat{y}\|$, $\|y' - \hat{y}\|$ and $\|y - y'\|$. The Triangle Inequality then gives us that

$$\|y' - \hat{y}\| \leq \|y - y'\| + \|y - \hat{y}\|.$$

Substituting using the three equations above, we have

$$\delta(\hat{h}_0, h') \leq \hat{R}(\hat{h}_0) + \hat{R}(h').$$

Since $h' \in S_\epsilon(\hat{h}_0)$, we have by the definition of the ϵ -level set that $\hat{R}(h') \leq \hat{R}(\hat{h}_0) + \epsilon$. We can then rewrite the above expression to yield

$$\delta(\hat{h}_0, h') \leq 2\hat{R}(\hat{h}_0) + \epsilon$$

Recall that $\delta_\epsilon := \max_{h' \in S_\epsilon(\hat{h}_0)} \delta(\hat{h}_0, h')$. Since each $h' \in S_\epsilon(\hat{h}_0)$ satisfies $\delta(\hat{h}_0, h') \leq 2\hat{R}(\hat{h}_0) + \epsilon$, we have the result that $\delta_\epsilon \leq 2\hat{R}(\hat{h}_0) + \epsilon$. $\square$

B Supporting Experimental Results

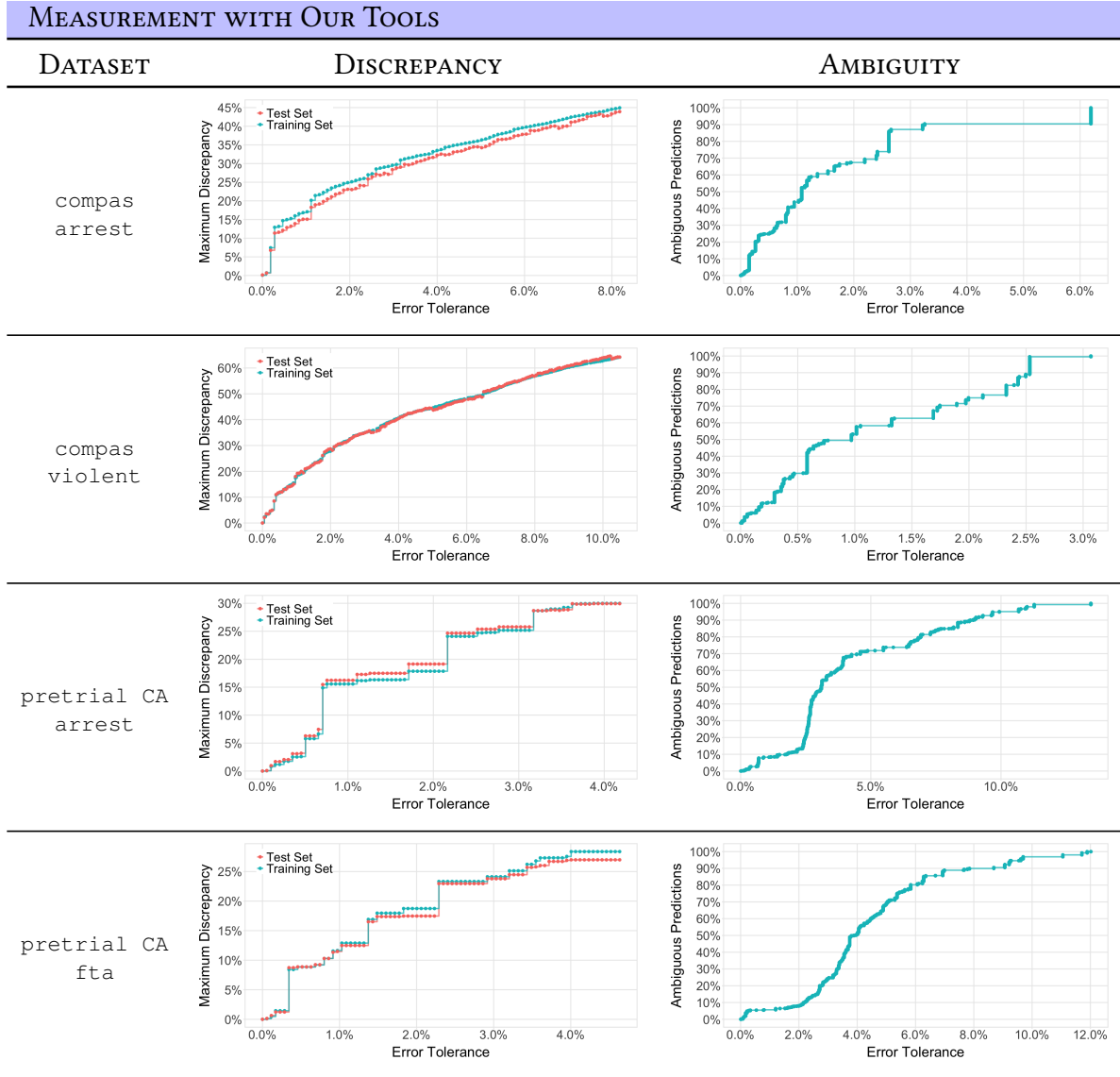


Fig. 4. Multiplicity profiles for the `compas` and `pretrial` datasets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (ambiguity) is shown on the right.

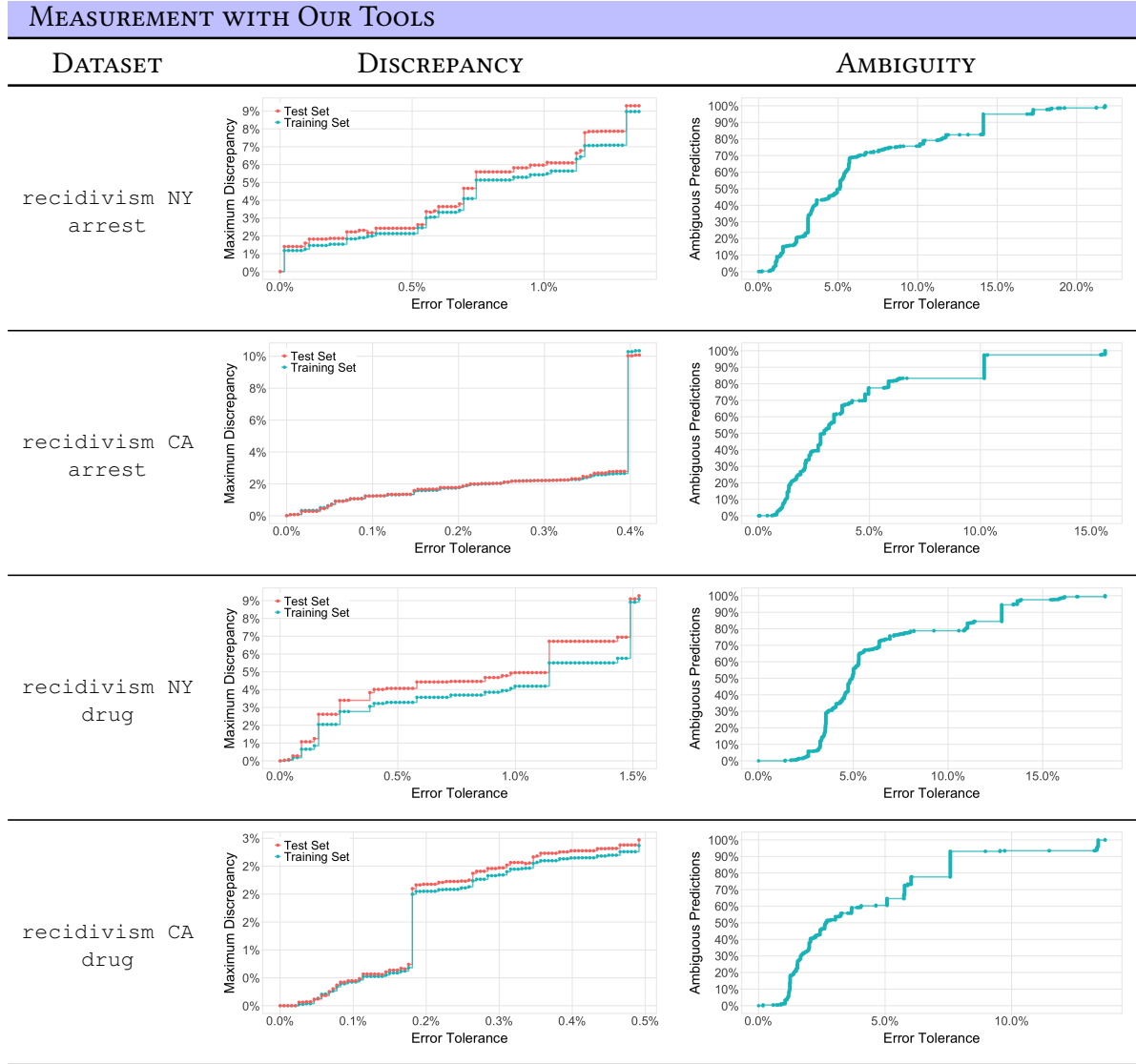


Fig. 5. Multiplicity profiles for the `recidivism` datasets. The maximum percent of conflicts between the optimal model and any model within the error tolerance (discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (ambiguity) is shown on the right.

MEASUREMENT WITH STANDARD TOOLS

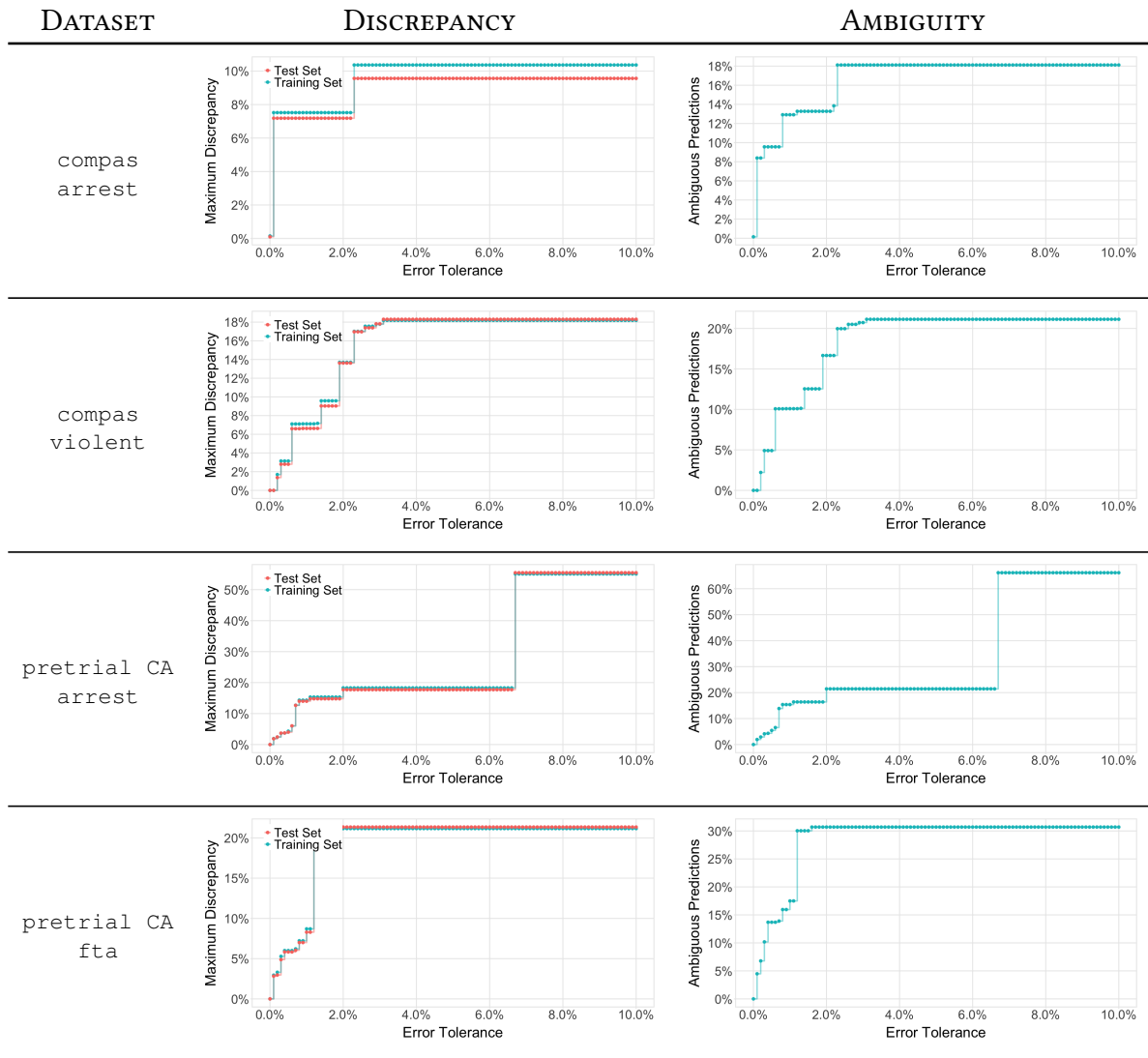


Fig. 6. Multiplicity profiles for the `compas` and `pretrial` datasets produced via ad hoc measurement. The maximum percent of conflicts between the optimal model and any model within the error tolerance (discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (ambiguity) is shown on the right.

MEASUREMENT WITH STANDARD TOOLS

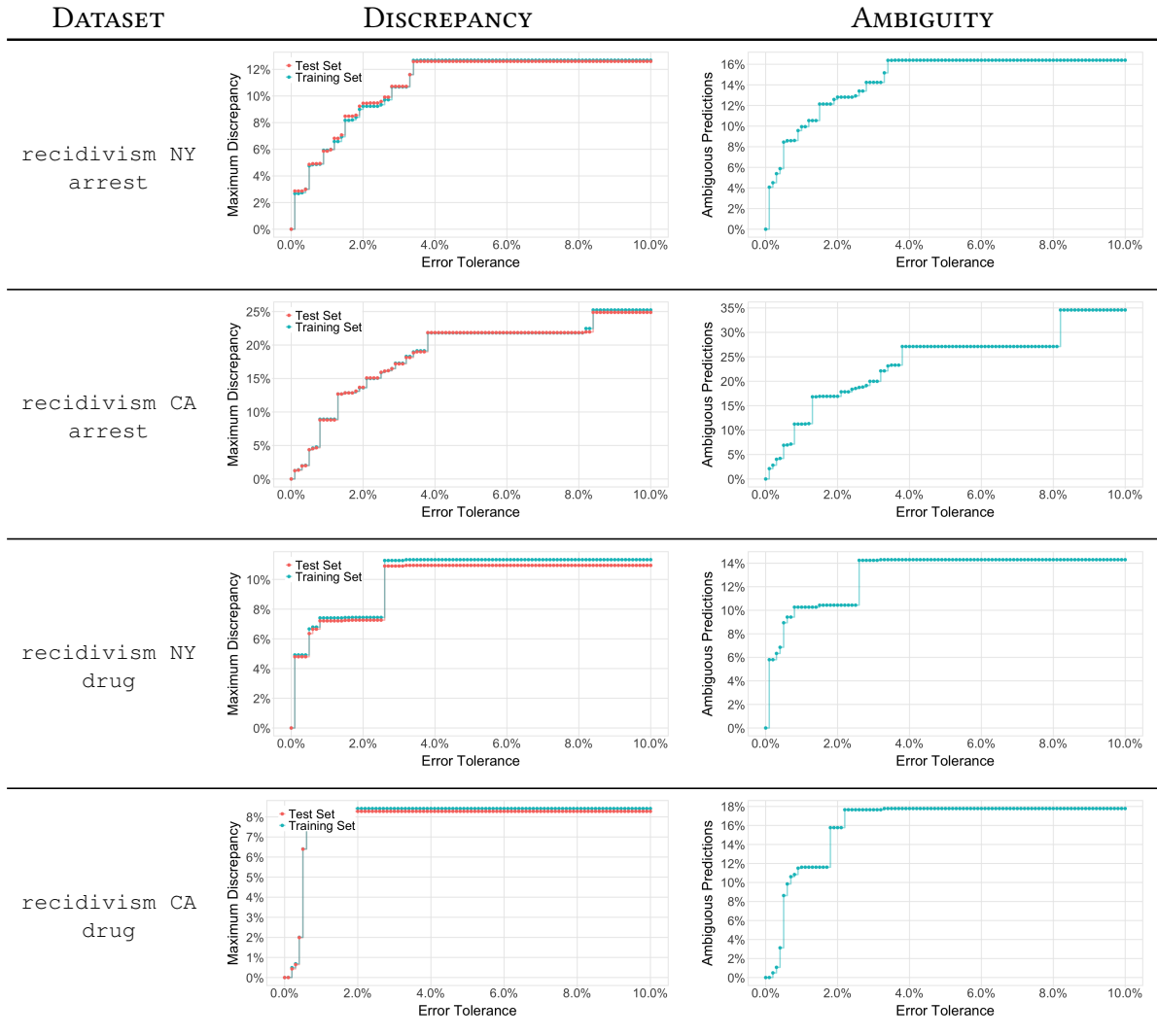


Fig. 7. Multiplicity profiles for the `recidivism` datasets produced via ad hoc measurement. The maximum percent of conflicts between the optimal model and any model within the error tolerance (discrepancy) is shown on the left. The percent of individuals who could have been given a different prediction by some model within the error tolerance (ambiguity) is shown on the right.